

Compile with: g++ -Wall -Werror main.cc

Run with ./a.out input_file.txt fcfs or ./a.out input_file.txt rr 3

Test Case: out of order PID:

4

0 0 12

1 2 4

3 4 2

2 3 1

```
jjoyce@LAPTOP-0GI2H9KF:~/CS4420/CS4420/assignment 05$ ./a.out input.txt fcfs
Process: 0 started at 0ms
Process: 0 ended at 12ms
Process: 1 started at 12ms
Process: 1 ended at 16ms
Process: 2 started at 16ms
Process: 2 ended at 17ms
Process: 3 started at 17ms
Process: 3 ended at 19ms
PID    Arrival Time    Start Time    End Time    Running Time    Waiting Time
0       0                0             12          12              0
1       2                12            16          4               10
2       3                16            17          1               13
3       4                17            19          2               13
Average wait time: 9
```

```
jjoyce@LAPTOP-0GI2H9KF:~/CS4420/CS4420/assignment 05$ ./a.out input.txt rr 5
Process: 0 started at 0ms
process: 0 is passed at 5ms
Process: 1 started at 5ms
Process: 1 ended at 9ms
Process: 2 started at 9ms
Process: 2 ended at 10ms
Process: 3 started at 10ms
Process: 3 ended at 12ms
process: 0 is continued at 12ms
process: 0 is passed at 17ms
process: 0 is continued at 17ms
Process: 0 ended at 19ms
PID    Arrival Time    Start Time    End Time    Running Time    Waiting Time
0       0                0             19          12              7
1       2                5              9           4               3
2       3                9             10          1               6
3       4                10            12          2               6
Average wait time: 5.5
```

```

jjoyce@LAPTOP-0GI2H9KF:~/CS4420/CS4420/assignment 05$ ./a.out input.txt sjf
Process: 0 started at 0ms
Process: 0 ended at 12ms
Process: 2 started at 12ms
Process: 2 ended at 13ms
Process: 3 started at 13ms
Process: 3 ended at 15ms
Process: 1 started at 15ms
Process: 1 ended at 19ms

```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	12	12	0
1	2	15	19	4	13
2	3	12	13	1	9
3	4	13	15	2	9

Average wait time: 7.75

Test Case Idle:

5

0 0 12

1 2 4

3 4 2

2 3 1

4 3 1 8

```

jjoyce@LAPTOP-0GI2H9KF:~/CS4420/CS4420/assignment 05$ ./a.out input.txt fcfs
Process: 0 started at 0ms
Process: 0 ended at 12ms
Process: 1 started at 12ms
Process: 1 ended at 16ms
Process: 2 started at 16ms
Process: 2 ended at 17ms
Process: 3 started at 17ms
Process: 3 ended at 19ms
4 is idle until 31ms
Process: 4 started at 31ms
Process: 4 ended at 39ms

```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	12	12	0
1	2	12	16	4	10
2	3	16	17	1	13
3	4	17	19	2	13
4	31	31	39	8	0

Average wait time: 7.2

```

jjjoyce@LAPTOP-0GI2H9KF:~/CS4420/CS4420/assignment 05$ ./a.out input.txt rr 5
Process: 0 started at 0ms
process: 0 is passed at 5ms
Process: 1 started at 5ms
Process: 1 ended at 9ms
Process: 2 started at 9ms
Process: 2 ended at 10ms
Process: 3 started at 10ms
Process: 3 ended at 12ms
Idle at 12ms
Process: 4 started at 13ms
process: 4 is passed at 18ms
process: 0 is continued at 18ms
process: 0 is passed at 23ms
Idle at 23ms
process: 4 is continued at 24ms
Process: 4 ended at 27ms
process: 0 is continued at 27ms
Process: 0 ended at 29ms
Idle at 29ms

```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	29	12	17
1	2	5	9	4	3
2	3	9	10	1	6
3	4	10	12	2	6
4	31	13	27	8	6

Average wait time: 7.6

```

jjjoyce@LAPTOP-0GI2H9KF:~/CS4420/CS4420/assignment 05$ ./a.out input.txt sjf
Process: 0 started at 0ms
Process: 0 ended at 12ms
Process: 2 started at 12ms
Process: 2 ended at 13ms
Process: 3 started at 13ms
Process: 3 ended at 15ms
Process: 1 started at 15ms
Process: 1 ended at 19ms
Process: 4 started at 31ms
Process: 4 ended at 39ms

```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	12	12	0
1	2	15	19	4	13
2	3	12	13	1	9
3	4	13	15	2	9
4	31	31	39	8	0

Average wait time: 6.2

Test Case all processes arrive at the same time

4

0 0 12

1 0 4

3 0 2

2 0 1

```

jjoyce@LAPTOP-0GI2H9KF:~/CS4420/CS4420/assignment 05$ ./a.out input.txt fcfs
Process: 0 started at 0ms
Process: 0 ended at 12ms
Process: 1 started at 12ms
Process: 1 ended at 16ms
Process: 3 started at 16ms
Process: 3 ended at 18ms
Process: 2 started at 18ms
Process: 2 ended at 19ms

```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	12	12	0
1	0	12	16	4	12
2	0	18	19	1	18
3	0	16	18	2	16

Average wait time: 11.5

```

jjoyce@LAPTOP-0GI2H9KF:~/CS4420/CS4420/assignment 05$ ./a.out input.txt rr 5
Process: 1 ended at 9ms
Process: 3 started at 9ms
Process: 3 ended at 11ms
Process: 2 started at 11ms
Process: 2 ended at 12ms
process: 0 is continued at 12ms
process: 0 is passed at 17ms
process: 0 is continued at 17ms
Process: 0 ended at 19ms

```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	0	19	12	7
1	0	5	9	4	5
2	0	11	12	1	11
3	0	9	11	2	9

Average wait time: 8

```

jjoyce@LAPTOP-0GI2H9KF:~/CS4420/CS4420/assignment 05$ ./a.out input.txt sjf
Process: 2 started at 0ms
Process: 2 ended at 1ms
Process: 3 started at 1ms
Process: 3 ended at 3ms
Process: 1 started at 3ms
Process: 1 ended at 7ms
Process: 0 started at 7ms
Process: 0 ended at 19ms

```

PID	Arrival Time	Start Time	End Time	Running Time	Waiting Time
0	0	7	19	12	7
1	0	3	7	4	3
2	0	0	1	1	0
3	0	1	3	2	1

Average wait time: 2.75